Local data of the Higman–Sims coset geometries

We acquire the reviewed classification of 86 coset geometries for the literal faithful group $G \cong$ HS, of order 44 352 000. The numbers in ranks two through six are $6, 30, 41, 8, 1$. The acquired action is flag-transitive. Its parabolics are denoted by G_i, $i \in I$, and $G_J = \bigcap_{j \in J} G_j$, with $G_\varnothing = G$. The Borel subgroup is $B = G_I$. All identifiers below refer to the fixed representatives in `full_fusion03/result.g`.

Theorem 1. *The retained finite calculations determine, for each of these 86 representatives, its type sizes, every flag stabilizer order, its Borel and panel orders, its node orders, and the orders and literal generators of its local action kernels. They also determine the two ordered diameters and girth of each of its 437 rank-two residues at a base flag. Finally, they determine the normalizer of the specified G action in the type-preserving automorphism group. For representatives 14 and 15 this is the full automorphism group, of respective orders* 44 352 000 *and* 88 704 000.

The assertion is a computation of the actual models. It does not assume agreement with any printed group name or diagram label. The historical supplement of 6 October 2003, attributed to Dimitri Leemans, is compared with these values in a separate source audit. No new classification or open-conjecture resolution is asserted here.

Stabilizers and local kernels

The objects of type i are the right cosets of G_i, so their number is $|G : G_i|$. The stabilizer of the standard flag of type J is exactly G_J. Flag transitivity gives $|G : G_J|$ flags of that type. Fixing the standard flag of cotype $\{i\}$ leaves $|G_{I\setminus\{i\}} : B|$ objects; consequently the node order is this index minus one.

Fix an object of type i. For $j \neq i$, its incident objects of type j form the transitive G_i-set with stabilizer $G_i \cap G_j$. The kernel on that set is the core of this intersection in G_i. An element acts trivially on the whole residue precisely when it acts trivially on every type. Thus the local kernel is

$$K_i = \bigcap_{j \neq i} \mathrm{Core}_{G_i}(G_i \cap G_j).$$

The calculation uses these actual intersections and cores. It saves generators of B, the panels, each K_i, and the original parabolics, as well as their orders. An order by itself is not a structural group identification.

An indexed construction of the incidence graphs

For a residue of cotype $\{i, j\}$ put $H = G_{I\setminus\{i,j\}}$, $P = H \cap G_i$, $Q = H \cap G_j$, and $D = P \cap Q = B$. Its two types are the cosets of P and Q in H. Take right transversals T_P, T_Q in H and a right transversal T_D of D in P. For $x \in T_P$, its full neighbour set is

$$\{Qux : u \in T_D\}. \tag{1}$$

Indeed a coset meeting Px contains some px, $p \in P$, and is therefore Qpx. Two such cosets agree precisely when $p_1 p_2^{-1} \in P \cap Q = D$. Thus (1) contains each neighbour exactly once. A

position lookup in T_Q gives the integer vertex label without constructing a large permutation action.

For the six rank-two geometries this construction produces 158,447 vertices and 537,835 edges in total. The completed invocation took 34.36 seconds. An earlier generic action construction reached its 120-second bound without returning a graph; that record is preserved. The new construction changes the indexing method, not the incidence relation. Independent small controls compare every edge with literal coset intersection.

For higher ranks, 425 graphs were obtained by transporting the base P-orbit in the action of H on Q-cosets. This is the same neighbour formula. Six graphs for representatives 14 and 15 are acquired from their earlier reviewed comparison. Together the 437 graphs have 187,005 vertices and 601,049 edges. The exact roster contains one graph for every unordered type pair of every representative. The reader checks every adjacency, both valencies, the two flag indices, and $|H : B|$ edges. These checks supplement the exact coset construction; they do not substitute matching counts for incidence.

Distances and the diagram parameters

Lemma 1. *In a finite connected bipartite graph containing a cycle, whose automorphism group is transitive on each part, breadth-first search at one vertex in each part determines both type diameters and the girth.*

Proof. The eccentricity is constant on each part by transitivity. Hence the largest distance found from its chosen vertex is that type's diameter. For a breadth-first tree rooted at v, a non-tree edge xy gives the number $d(v, x) + d(v, y) + 1$. Its tree paths and the edge contain a cycle of length at most this number, so every such number is at least the girth. Choose a shortest cycle and move a vertex of it to v using transitivity on the appropriate part. Its length is even, say $2m$. There is no shorter path between vertices of this cycle than the shorter arc, since such a path would give a shorter cycle. The vertex opposite v therefore has distance m, and its two cycle neighbours have distance $m - 1$. At least one of these two edges is not a tree edge and gives $2m$. The minimum recorded number is consequently the girth. This argument applies to either part. □

The acquired coset action is transitive on each part. All valencies are at least two, so these finite graphs contain cycles and the lemma applies. The two searches must agree on the girth. The diagram triple is the left type diameter, half the girth, and the right type diameter. A digon is checked as a complete bipartite graph. A complete-graph residue is recognized only when one type consists of all distinct two-subsets of the other, with exactly that incidence.

Normalizers and the two remaining ambiguous profiles

Let $N \cong \mathrm{Aut}(G)$ be the acquired faithful automorphism action on G, and put $S = \bigcap_i N_N(G_i)$. Let M be the normalizer of the specified G action in the type-preserving geometry automorphism group. The chamber stabilizer M_Φ maps by conjugation into S. This map is injective: an element fixing Φ and commuting with G fixes every chamber, because G is chamber-transitive, and hence fixes every object. Conversely each automorphism in S induces the chamber-fixing coset map $G_i x \mapsto G_i \alpha(x)$. Thus $M_\Phi \cong S$. Since $G \leq M$ is chamber-transitive,

$$|M| = |G : B|\,|S|.$$

This formula computes a normalizer; it does not in general prove that all geometry automorphisms normalize the given copy of G.

For representatives 14 and 15, that last assertion is already paid by the reviewed shadow argument. Each defines a graph on its 1100-point type; restriction to that type is faithful, and the derived subgroup of the full shadow-graph group is precisely the specified G. Every geometry automorphism therefore normalizes G. Their Borel orders are both 24, while the calculated orders of S are respectively 24 and 48. The displayed formula gives the two full orders stated above. All three type sizes are different, so their correlation groups preserve types as well. For representative 15 the resulting group is the full shadow-graph group, which is isomorphic to $\mathrm{Aut}(G)$ by its faithful conjugation action. Representative 14 has full group equal to the specified G itself. To justify the faithful conjugation assertion, the calculation also checks $S \cap G = B$ in both cases. An automorphism commuting with G takes the base chamber to Φ^g for some $g \in G$; equality of the object stabilizers forces $g \in S \cap G = B$. It therefore fixes the base chamber and, by transitivity, the entire geometry. The conjugation kernel is trivial.

Attribution and computational boundary. The group model, complete family classification, full automorphism action and faithful shadow argument are acquired reviewed results. The general arguments here are elementary coset and group-action facts; their explicit derivations, numerical applications and validation programs are supplied here. Exact GAP permutation-group algorithms remain computational premises. No historical source label, full automorphism claim for the other 84 models, abstract identification of local kernels, Lean verification, measurement or whole-manuscript completion follows from this component alone.